

Anna Tang

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SKILLS

Python, SQL, R, Data Analytics [Pandas, NumPy, SciPy, statsmodels], Data Visualization [Matplotlib, Plotly, Seaborn, Geopandas, Shiny], Machine Learning [SciKit-Learn, TensorFlow, Keras, PyTorch], Database Management [PostgreSQL, BigQuery], Tools [Git, Docker, Jupyter], Microsoft Office, Amazon Web Services, Azure, Snowflake

EDUCATION

Master of Science in Data Science | University of Colorado Boulder | 2024 | 3.96 GPA

Data Science Career Track | Springboard | 2020

Bachelor of Arts in Mathematics | The University of North Carolina at Chapel Hill | 2018

EXPERIENCE

Graduate Student | University of Colorado Boulder | August 2023 - Present

- Optimized sentiment analysis and topic modeling NLP models using Azure ML and TensorFlow, using transformer-based architectures and customized LLMs to extract user insights and highlight keywords. Improved model accuracy by 15% through hyper-parameter tuning and topic modeling optimization.
- Managed end-to-end NLP pipelines using Keras, PyTorch, and BERT, DistilBERT, and RoBERTa for comprehensive text analysis, allowing improved unstructured data interpretation across various document formats.
- Automated data collection and validation workflows using Python and SQLAlchemy, and Azure ML Studio. Included rigorous data quality checks and preprocessing techniques streamlining multi-source data integration and enabling predictive analytics reporting.

Data Scientist | Whiteboard Coordinator | June 2022 - July 2024

- Created and deployed an interactive analytics dashboards using Python, SQL, and AWS QuickSight with real-time monitoring and predictive analytics features to forecast usage patterns, resulting in a 30% reduction for manual reporting workflows.
- Designed and implemented a medallion architecture PostgreSQL database using automated ETL pipelines and performance optimization techniques, resulting in 60% improvement of application response times.
- Developed an automated model drift detection system using SageMaker Model Monitor and custom Python scripts with live alerts for performance drift and a guide for rigorous user testing, resulting in a 30% improvement in model reliability.

Machine Learning Engineer | Omdena | Freelance | June 2020 - April 2022

- Developed an end-to-end ETL pipeline on AWS to analyze Hashimoto's disease patterns, using Amazon RDS for structured data storage.
- Deployed clustering algorithms through SageMaker with automated model retraining, creating a symptom identification system that generates personalized management recommendations through AWS Lambda and API Gateway endpoints.

Data Scientist | Kamsa | June 2021 - December 2021

- Developed an automated budgeting forecast solution using AWS Lambda and SageMaker to implement ARIMA time-series analysis and gradient boosting models to generate data-driven financial insights, improving data processing times by 30%.
- Collaborated with engineering and DevOps teams to deploy a scalable analytics application on AWS ECS with interactive analytics reporting on QuickSight dashboards, reducing manual processes by 20%.

Data Analyst | CALIBRE Systems | May 2020 - June 2021

- Created an interactive map application to visualize socioeconomic demographics surrounding military installations using python, geopandas, and QGIS with custom layouts and user-friendly customization features.
- Increased accuracy metrics by 26% for a budget requirements regression model using PCA, modified z-scores, and clustering analysis methods with logistic and lasso regression models.
- Supported an NLP document classification and topic modeling project to analyze, sort, and tag documents based on contract guidelines and requirements.

Media Analyst | Duke University | November 2018 - November 2019

- Increased media engagement by 20% using target market research to create successful marketing campaigns with detailed timelines, actionable KPIs, and ongoing analysis of key metrics to identify emerging trends.
- Developed web and media dashboards using Tableau highlighting key metrics and new insights on traffic patterns, creating new customer segmentation methods and research outreach strategies.
- Created automated daily traffic analysis reports using python and SQL scripts, saving ~15 hours per week in manual report generations.

Web Developer | National Institute of Environmental Health Sciences | May 2016 - October 2018

- Boosted web traffic by 40% through full stack development efforts using python, HTML, CSS, and JavaScript to enhance site navigation tools and optimize social media plugins and a comprehensive site redesign.
- Managed a web analytics database and wrote custom SQL queries for data extraction, traffic pattern analysis, and ad-hoc report generation for stakeholders. Automated deliverables using Excel to include visualizations and dashboards, informing clients with clear and actionable data insights.
- Created dashboards in Tableau to assist stakeholders with monitoring user activities and tracking site performance with KPIs.